

Exhibit CC-142

DDR5 SDRAM RDIMM Addendum

MTC40F204WS1RC – 96GB 24Gb Die Revision B

Features

Information provided here is in addition to or supercedes information provided in the Micron DDR5 RDIMM Core data sheet.

- DDR5 functionality and operations supported as defined in the component data sheet
- Features and specifications defined in the Micron DDR5 RDIMM core data sheet
- 288-pin, DDR5 registered dual in-line memory module (DDR5 RDIMM)
- Fast data transfer rate: PC5-4800, PC5-5600, PC5-6400
- 96GB (12Gig x 80)
- Dual-rank
- 32 internal banks; 8 groups of 4 banks each

Figure 1: 288-Pin DDR5 RDIMM (R/C-A0)

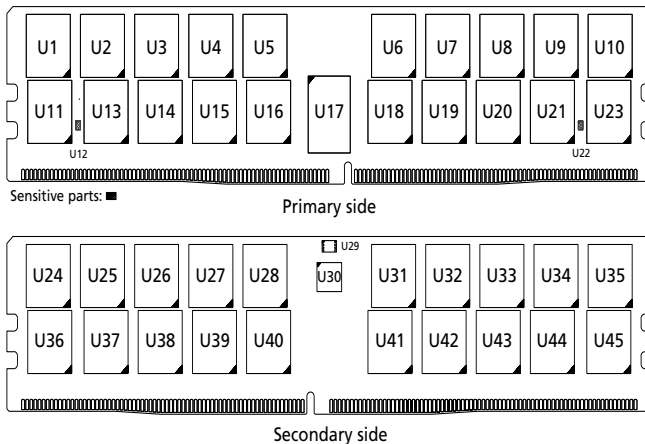
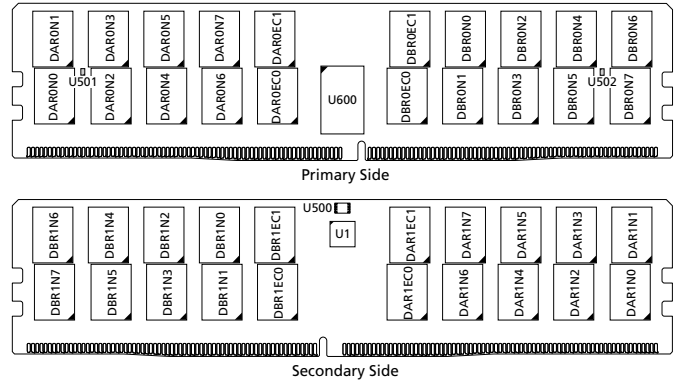


Figure 2: 288-Pin DDR5 RDIMM (R/C-A1)



Options

- Operating temperature
 - Commercial ($0^{\circ}\text{C} \leq T_{\text{OPER}} \leq 95^{\circ}\text{C}$)
- Frequency/CAS latency
 - 0.416ns @ CL = 40 (DDR5-4800)
 - 0.357ns @ CL = 46 (DDR5-5600)
 - 0.312ns @ CL = 52 (DDR5-6400)

Marking

C
48B
56B
64B



96GB (x80, ECC, DR) 288-Pin DDR5 RDIMM Features

Table 1: Addressing

Parameter	96GB
Row address	96K (R0-R16) ¹
Column address	2K (C0-C10)
Device bank group address	8 (BG0-BG2)
Device bank address per bank group	4 (BA0-BA1)
Device configuration	24Gb (6Gb x 4), 32 banks
Module rank address	2 (CS0_n, CS1_n)

Notes: 1. For non-binary densities, a quarter of the row address space is invalid. When the MSB address bit is HIGH, the MSB-1 address shall be LOW.

Table 2: Part Numbers and Timing Parameters – 96GB Modules

Base device: MT60B6G4,¹ 24Gb DDR5 SDRAM Die Revision B

Part Number	Module Density	Configuration	Module Bandwidth	Memory Clock/ Data Rate	Clock Cycles (CL _n RCD _n RP)	Designation ²
MTC40F204WS1RC48BB1	96GB	12Gb x 80 (EC8)	38.4 GB/s	0.416ns/4800 MT/s	40-39-39	Production
MTC40F204WS1RC56BB1	96GB	12Gb x 80 (EC8)	44.8 GB/s	0.357ns/5600 MT/s	46-45-45	Preliminary
MTC40F204WS1RC64BB1	96GB	12Gb x 80 (EC8)	51.2 GB/s	0.312ns/6400 MT/s	52-52-52	Advance

Notes: 1. The data sheet for the base device can be found on [micron.com](https://www.micron.com).

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DQ Map

Table 3: Component-to-Module DQ Map (R/C A-0)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U1	0	7A	161	U2	0	15A	172
	1	4A	14		1	12A	25
	2	6A	159		2	14A	170
	3	5A	16		3	13A	27
U3	0	23A	183	U4	0	31A	194
	1	20A	36		1	28A	47
	2	22A	181		2	30A	192
	3	21A	38		3	29A	49
U5	0	CB7A	205	U6	0	CB7B	236
	1	CB4A	58		1	CB4B	89
	2	CB6A	203		2	CB6B	234
	3	CB5A	60		3	CB5B	91
U7	0	3B	247	U8	0	11B	258
	1	0B	100		1	8B	111
	2	2B	245		2	10B	256
	3	1B	102		3	9B	113
U9	0	19B	269	U10	0	27B	280
	1	16B	122		1	24B	133
	2	18B	267		2	26B	278
	3	17B	124		3	25B	135
U11	0	3A	154	U13	0	11A	165
	1	0A	7		1	8A	18
	2	2A	152		2	10A	163
	3	1A	9		3	9A	20
U14	0	19A	176	U15	0	27A	187
	1	16A	29		1	24A	40
	2	18A	174		2	26A	185
	3	17A	31		3	25A	42
U16	0	CB3A	198	U18	0	CB3B	243
	1	CB0A	51		1	CB0B	96
	2	CB2A	196		2	CB2B	241
	3	CB1A	53		3	CB1B	98



Table 3: Component-to-Module DQ Map (R/C A-0) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U19	0	7B	254	U20	0	15B	265
	1	4B	107		1	12B	118
	2	6B	252		2	14B	263
	3	5B	109		3	13B	120
U21	0	23B	276	U23	0	31B	287
	1	20B	129		1	28B	140
	2	22B	274		2	30B	285
	3	21B	131		3	29B	142
U24	0	24B	133	U25	0	16B	122
	1	27B	280		1	19B	269
	2	25B	135		2	17B	124
	3	26B	278		3	18B	267
U26	0	8B	111	U27	0	0B	100
	1	11B	258		1	3B	247
	2	9B	113		2	1B	102
	3	10B	256		3	2B	245
U28	0	CB4B	89	U31	0	CB4A	58
	1	CB7B	236		1	CB7A	205
	2	CB5B	91		2	CB5A	60
	3	CB6B	234		3	CB6A	203
U32	0	28A	47	U33	0	20A	36
	1	31A	194		1	23A	183
	2	29A	49		2	21A	38
	3	30A	192		3	22A	181
U34	0	12A	25	U35	0	4A	14
	1	15A	172		1	7A	161
	2	13A	27		2	5A	16
	3	14A	170		3	6A	159
U36	0	28B	140	U37	0	20B	129
	1	31B	287		1	23B	276
	2	29B	142		2	21B	131
	3	30B	285		3	22B	274
U38	0	12B	118	U39	0	4B	107
	1	15B	265		1	7B	254
	2	13B	120		2	5B	109
	3	14B	263		3	6B	252

Table 3: Component-to-Module DQ Map (R/C A-0) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U40	0	CB0B	96	U41	0	CB0A	51
	1	CB3B	243		1	CB3A	198
	2	CB1B	98		2	CB1A	53
	3	CB2B	241		3	CB2A	196
U42	0	24A	40	U43	0	16A	29
	1	27A	187		1	19A	176
	2	25A	42		2	17A	31
	3	26A	185		3	18A	174
U44	0	8A	18	U45	0	0A	7
	1	11A	165		1	3A	154
	2	9A	20		2	1A	9
	3	10A	163		3	2A	152

Table 4: Component-to-Module DQ Map (R/C A-1)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DAR0EC0	0	CB3A	198	DAR0EC1	0	CB7A	205
	1	CB0A	51		1	CB4A	58
	2	CB2A	196		2	CB6A	203
	3	CB1A	53		3	CB5A	60
DAR0N0	0	3A	154	DAR0N1	0	7A	161
	1	0A	7		1	4A	14
	2	2A	152		2	6A	159
	3	1A	9		3	5A	16
DAR0N2	0	11A	165	DAR0N3	0	15A	172
	1	8A	18		1	12A	25
	2	10A	163		2	14A	170
	3	9A	20		3	13A	27
DAR0N4	0	19A	176	DAR0N5	0	23A	183
	1	16A	29		1	20A	36
	2	18A	174		2	22A	181
	3	17A	31		3	21A	38



Table 4: Component-to-Module DQ Map (R/C A-1) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DAR0N6	0	27A	187	DAR0N7	0	31A	194
	1	24A	40		1	28A	47
	2	26A	185		2	30A	192
	3	25A	42		3	29A	49
DAR1EC0	0	CB0A	51	DAR1EC1	0	CB4A	58
	1	CB3A	198		1	CB7A	205
	2	CB1A	53		2	CB5A	60
	3	CB2A	196		3	CB6A	203
DAR1N0	0	0A	7	DAR1N1	0	4A	25
	1	3A	154		1	7A	172
	2	1A	9		2	5A	27
	3	2A	152		3	6A	170
DAR1N2	0	8A	18	DAR1N3	0	12A	25
	1	11A	165		1	15A	172
	2	9A	20		2	13A	27
	3	10A	163		3	14A	170
DAR1N4	0	16A	29	DAR1N5	0	20A	36
	1	19A	176		1	23A	183
	2	17A	31		2	21A	38
	3	18A	174		3	22A	181
DAR1N6	0	24A	40	DAR1N7	0	28A	47
	1	27A	187		1	31A	194
	2	25A	42		2	29A	49
	3	26A	185		3	30A	192
DBR0EC0	0	CB3B	243	DBR0EC1	0	CB7B	236
	1	CB0B	96		1	CB4B	89
	2	CB2B	241		2	CB6B	234
	3	CB1B	98		3	CB5B	91
DBR0N0	0	3B	247	DBR0N1	0	7B	254
	1	0B	100		1	4B	107
	2	2B	245		2	6B	252
	3	1B	102		3	5B	109
DBR0N2	0	11B	258	DBR0N3	0	15B	265
	1	8B	111		1	12B	118
	2	10B	256		2	14B	263
	3	9B	113		3	13B	120

Table 4: Component-to-Module DQ Map (R/C A-1) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DBR0N4	0	19B	269	DBR0N5	0	23B	276
	1	16B	122		1	20B	129
	2	18B	267		2	22B	274
	3	17B	124		3	21B	131
DBR0N6	0	27B	280	DBR0N7	0	31B	287
	1	24B	133		1	28B	140
	2	26B	278		2	30B	285
	3	25B	135		3	29B	142
DBR1EC0	0	CB0B	96	DBR1EC1	0	CB4B	89
	1	CB3B	243		1	CB7B	236
	2	CB1B	98		2	CB5B	91
	3	CB2B	241		3	CB6B	234
DBR1N0	0	0B	100	DBR1N1	0	4B	107
	1	3B	247		1	7B	254
	2	1B	102		2	5B	109
	3	2B	245		3	6B	252
DBR1N2	0	8B	111	DBR1N3	0	12B	118
	1	11B	258		1	15B	265
	2	9B	113		2	13B	120
	3	10B	256		3	14B	263
DBR1N4	0	16B	122	DBR1N5	0	20B	129
	1	19B	269		1	23B	276
	2	17B	124		2	21B	131
	3	18B	267		3	22B	274
DBR1N6	0	24B	133	DBR1N7	0	28B	140
	1	27B	280		1	31B	287
	2	25B	135		2	29B	142
	3	26B	278		3	30B	285



I_{DD} Specifications

Table 5: DDR5 I_{DD} Specifications and Conditions – 96GB (Die Revision B)

Module I_{DD} is based on PMIC VIN_BULK 12V input current and typical operating range of temperature. Each I_{DD} parameter includes PMIC efficiency, RCD current and all DRAM current on all supplies (V_{DD}, V_{DDQ}, and V_{PP}).

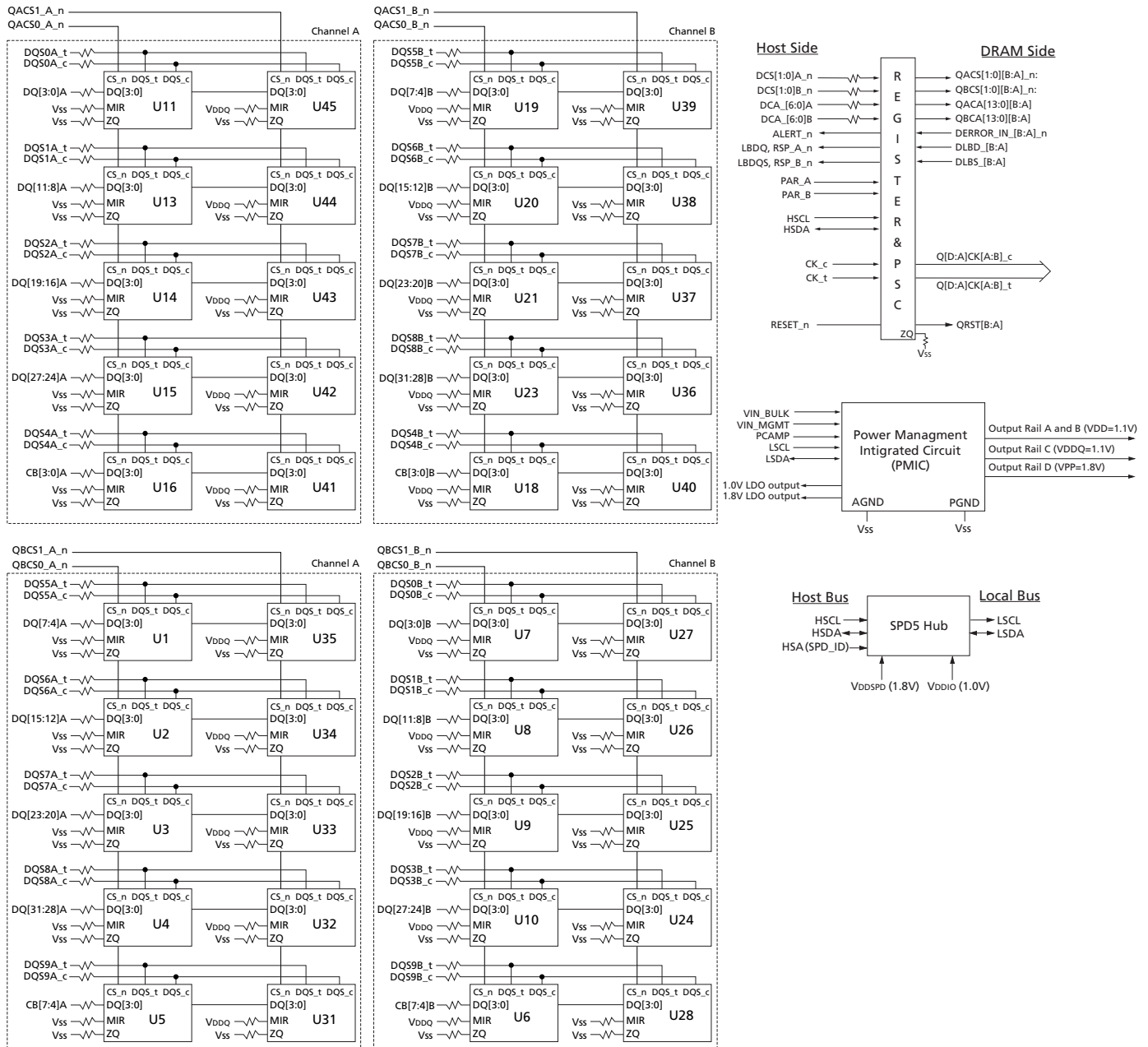
Parameter	Symbol	4800	5600	6400	Units
Operating one bank ACTIVATE-PRECHARGE current	I _{DD0} ¹	443	TBD	TBD	mA
Operating four bank ACTIVATE-PRECHARGE current	I _{DD0F} ¹	520	TBD	TBD	mA
Precharge standby current	I _{DD2N} ²	414	TBD	TBD	mA
Precharge standby non-target command	I _{DD2NT} ¹	494	TBD	TBD	mA
Precharge power-down current	I _{DD2P} ²	409	TBD	TBD	mA
Active standby current	I _{DD3N} ²	470	TBD	TBD	mA
Active power-down current	I _{DD3P} ²	465	TBD	TBD	mA
Operating burst read current	I _{DD4R} ¹	901	TBD	TBD	mA
Operating burst write current	I _{DD4W} ¹	1058	TBD	TBD	mA
Operating burst write with write CRC current	I _{DD4WC} ¹	1055	TBD	TBD	mA
Burst refresh (normal refresh mode) current	I _{DD5B} ¹	884	TBD	TBD	mA
Burst refresh (fine granularity refresh mode) current	I _{DD5F} ¹	858	TBD	TBD	mA
Burst refresh (same bank refresh mode) current	I _{DD5C} ¹	554	TBD	TBD	mA
Self refresh current	I _{DD6N} ²	345	TBD	TBD	mA
Operating bank interleave read current	I _{DD7} ¹	1220	TBD	TBD	mA
Maximum power saving deep power down mode current	I _{DD8} ²	328	TBD	TBD	mA

Notes: 1. One module rank in this I_{DD}/I_{DDQ}/I_{PP} condition, the other rank in I_{DD2N}/I_{DDQ2N}/I_{PP2N}.

2. Both ranks in this I_{DD}/I_{DDQ}/I_{PP} condition.

Functional Block Diagram

Figure 3: Functional Block Diagram



- Notes:
1. The ZQ ball on each DDR5 component is connected to an external $240\Omega \pm 1\%$ resistor that is tied to ground. It is used for the calibration of the component's ODT and output driver.
 2. Functional block diagram is for reference only.



Revision History

Rev. C – 01/2023

- Added IDD limits for 48B, and changed part status for 48B to "Production"
- Removed Micron Confidential marking
- Changed data sheet status to Production

Rev. B – 11/2022

- Added MIR state to Functional Block Diagram
- Added PCB drawing to Features page
- Added values for R/C A-1 DQ map

Rev. A – 04/2022

- Preliminary Release

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